Bayesian Analysis of Daily German and UK Stock Market Returns

EuStockMarkets (1991-1998) | Bayesian normal and Student-t regression | 1,859 daily return observations

Executive summary

This study evaluates the contemporaneous relationship between German DAX and UK FTSE daily log returns using 1,859 observations from the public R EuStockMarkets dataset. A Bayesian linear regression with normal errors was compared with a robust Student-t alternative, both fitted using JAGS. The estimated DAX slope was approximately 0.494 in both models; under the Student-t model its 95% credible interval was [0.4666, 0.5209], with all 30,000 sampled posterior slopes positive. The Student-t model achieved a lower deviance information criterion (DIC 3324.5 versus 3453.8), and reproduced the observed extreme-residual count more plausibly. Both models nevertheless exhibited residual serial dependence and dependence in squared residuals. The findings support strong same-day association, not causation or next-day predictability.

Introduction

The study asks (1) whether DAX and FTSE daily returns are positively associated, (2) how large the conditional mean association is, and (3) whether a heavy-tailed error model improves the representation of large market moves. Let x_i and y_i denote same-day DAX and FTSE returns, expressed in percentage points. The principal inferential quantity is the posterior distribution of the slope β, particularly Pr(β > 0 | data). Since markets react to common news and closing times need not coincide, the regression is descriptive rather than causal.

Data

The EuStockMarkets dataset distributed with base R contains 1,860 business-time closing-price observations for DAX, SMI, CAC, and FTSE from 1991-1998 (source: Erste Bank AG; see References). This analysis uses DAX and FTSE only. Data were loaded with **data("EuStockMarkets")**. No missing values were detected in any of the four series. The price levels were converted to 100 × successive log-price differences, yielding 1,859 synchronized daily returns for each market. The principal preparation challenge was avoiding a misleading regression on trending index levels rather than returns.

Summary statistic	DAX (%)	FTSE (%)
Mean daily log return	0.0652	0.0432
Median	0.0473	0.0080
Minimum / maximum	−9.6277 / 5.0760	−4.1399 / 5.4396
Pearson correlation	0.6395	(between DAX and FTSE)

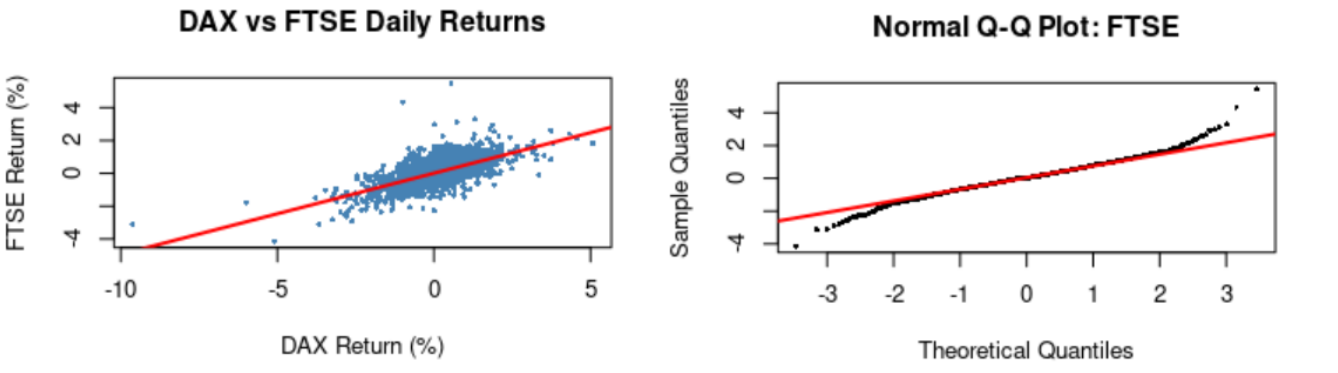


Figure 1. Exploratory plots from R: a positive same-day return association (left) and non-normal FTSE return tails in the marginal Q-Q plot (right).

Model specification and checking

Model

Both models use a linear conditional mean $\mu_i = \alpha + \beta x_i$. Conditional on their parameters, observations are treated as independent; this assumption is assessed below. Parameter priors are independent, with returns measured in percentage points.

Model 1 (normal errors): $y_i \mid \alpha, \beta, \sigma \sim \text{Normal}(\mu_i, \sigma^2)$, for i = 1, ..., 1859.

Model 2 (Student-t errors): $y_i \mid \alpha, \beta, \sigma, \nu \sim$ Student-t(location μ_i, scale σ, df ν), with $\nu = 2 + \eta$.

Priors, shared unless noted: $\alpha \sim \text{Normal}(0, 1^2)$; $\beta \sim \text{Normal}(0, 2^2)$; $\sigma \sim \text{Uniform}(0, 10)$; $\eta \sim \text{Exponential}(\text{rate} = 0.1)$, Model 2 only.

The centered, comparatively diffuse priors allow either sign of β, while limiting implausibly extreme coefficients. A positive, broad prior is used for residual scale. The shifted exponential prior enforces $\nu > 2$ (finite Student-t variance) while allowing heavy tails. In JAGS, the normal prior precisions are 1 and 0.25, respectively, and the likelihood uses precision $1/\sigma^2$. For the Student-t likelihood, σ is the scale, not its standard deviation.

Estimation: Models were fitted in R with rjags and coda, using three chains, 1,000 adaptation iterations, 5,000 burn-in iterations per chain and 10,000 retained draws per chain (30,000 total), seed 123. The monitored parameters were α, β and σ, plus ν for Model 2. DIC used 5,000 further JAGS iterations for each model.

Model checking and iteration

Convergence checks were reassuring: Gelman-Rubin potential scale reduction factors were reported as 1.00 for all parameters. Effective sample sizes were at least 18,047 for Model 1 and at least 6,865 for Model 2. For the normal model, the fitted residuals had kurtosis 8.28; 11 residuals exceeded three posterior mean normal error standard deviations, compared with approximately 5.02 expected under Gaussian errors. The residual Q-Q plot shows marked tail deviations, motivating Model 2.

Ljung-Box tests at 20 lags reject no serial correlation for both models. For Model 1, residual $p = 2.402 \times 10^{-6}$ and squared-residual $p = 8.751 \times 10^{-6}$; for Model 2, the corresponding values are 2.396×10^{-6} and 7.781×10^{-6}. These results indicate remaining serial dependence and are consistent with volatility clustering; merely replacing normal errors by Student-t errors does not remove temporal dependence.

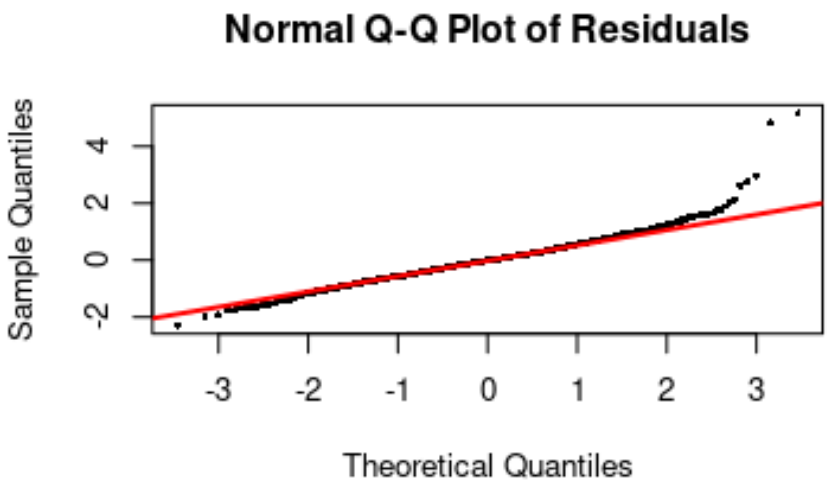


Figure 2. Normal-regression residual Q-Q plot: substantial departures from normality occur in the tails.

Results, conclusions and reproducibility

Results

Parameter / metric	Normal	Student-t
Slope β: posterior mean	0.49399	0.49382
β: 95% credible interval	[0.46725, 0.52094]	[0.46660, 0.52091]
Pr(β > 0 | data), MCMC estimate	1.000	1.000
Intercept α: posterior mean	0.01081	−0.00202
Error parameter σ: mean	0.61233 (SD)	0.51183 (scale)
Degrees of freedom ν: mean (95% CrI)	—	7.303 [5.503, 9.680]
DIC (lower preferred)	3453.81	3324.52

The near-identical slopes imply that the central association estimate is robust to the choice between normal and Student-t error models. A one-percentage-point higher DAX daily return is associated with approximately a 0.494-percentage-point higher conditional mean FTSE return on the same day. All sampled posterior slopes were positive (Monte Carlo estimate 1.000; this is not absolute certainty). The Student-t model yields a DIC lower by 129.29, favoring its trade-off between fit and complexity.

Posterior predictive check. For each model, 500 posterior parameter draws were used to simulate 1,859 independent errors per draw. The diagnostic counted |error| > 3 × 0.61233 = 1.837 percentage points, using the same fixed cutoff in both models. This is a check of marginal tail behavior, not of time dependence.

Extreme residual count	Observed	Normal predictive	Student-t predictive
Count / predictive mean	11	4.956	15.814
95% predictive interval	—	[1, 10]	[8, 25.53]

The observed count lies outside the normal model's simulated 95% interval but inside the Student-t model's. The latter still overpredicts the count on average; thus it offers an improvement, not a perfect tail fit.

Conclusions

The data provide strong posterior support for a positive contemporaneous DAX-FTSE return association. The preferred Student-t model estimates β = 0.494 (95% CrI 0.467-0.521) and better accommodates extreme daily moves than the normal regression. However, statistically significant residual and squared-residual autocorrelation indicate that both independent, constant-scale models remain incomplete. Credible intervals may be too narrow under temporal dependence; alternative specifications with serially correlated errors or time-varying volatility would be worthwhile. These estimates describe historical co-movement in the 1991-1998 sample and cannot establish a causal effect or useful next-day return forecasts.

Reproducibility and sources

Data preparation: data("EuStockMarkets"); returns <- as.data.frame(100 * diff(log(EuStockMarkets))). Set x = returns$DAX and y = returns$FTSE (the 1,859 paired observations). Run the two JAGS specifications and sampling settings described above; compute DIC with dic.samples(type="pD"). No raw source data were redistributed.

Sources: R Core Team, datasets::EuStockMarkets, R Documentation, https://stat.ethz.ch/R-manual/R-devel/library/datasets/html/EuStockMarkets.html (original data: Erste Bank AG, Vienna). JAGS 4.3 user manual, Student-t distribution parameterization. Figures are from the reported R analysis.